

Infrasonic Observations at Antarctica

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ABSTRACT

This is a report on microbarograph observations at Antarctica with the Indian Scientific Expedition of 1982-83. The general features are described and specific effects of magnetic disturbances as compared to observations taken in India are highlighted. Some earthquake and explosion effects are also indicated.

INTRODUCTION

The Second Indian Scientific Expedition Team to Antarctica (1982-83) set up a base camp at 69°59'S, 11°55'E and carried out various experiments with ground based instruments for a period of nearly 40 days from 8 Jan. to 18 Feb., 1983. One microbarograph was set up by NPL group primarily for studying magnetic disturbance effects near the auroral zone, for comparing the earlier findings with microbarographs that are in operation for several years at NPL, New Delhi and at NGRI, Hyderabad. The microbarographs at Delhi, Hyderabad and Antarctica are of identical type, the elements being condenser microphone type, with a frequency response of 0.001 to 2 Hz.

In this communication, we are reporting the general features of infrasonics that are observed at the Antarctica as compared to what are observed at the Indian stations. The infrasonics correlated with magnetic storms and bays (recorded by the Hyderabad Magnetic Observatory) are specially studied. Pressure fluctuations apparently associated with earthquakes/explosions of ocean bed origin are also illustrated.

The observations reported here are our initial efforts in the Second Antarctica Expedition and are for a limited period of 40 days only. This study justifies the need for future studies over longer periods of time.

INFRASONIC RECORDS ON NORMAL DAYS

Antarctica is an icy continent and the normal daytime temperature during summer is around 0 to -6°C (Night time temperature is around -6° to -12°C). During later part of the period of observation, the sunset occurred around 2200 hrs U.T. and the sunrise around 0200 hrs U.T. A representative daytime record is shown in Fig. 1. The daytime variation is within ± 200 microbars whereas for a low latitude station, it is of the order of ± 1000 microbars (Venkatachari and Bhartendu, 1979). Since the insolation received from the sun is small, convective heating and the consequent daytime turbulence are not as intense as in the tropics. Daytime and night time records are not very different, except that the amplitudes are sometimes reduced during night.

INFRASONICS AND MAGNETIC ACTIVITY

During magnetic storms, infrasonic pressure variations in Antarctica are found to be quite agitated as compared to mid-latitude in conformity with the measurements made in the Arctic region by earlier workers (Campbell, 1967).

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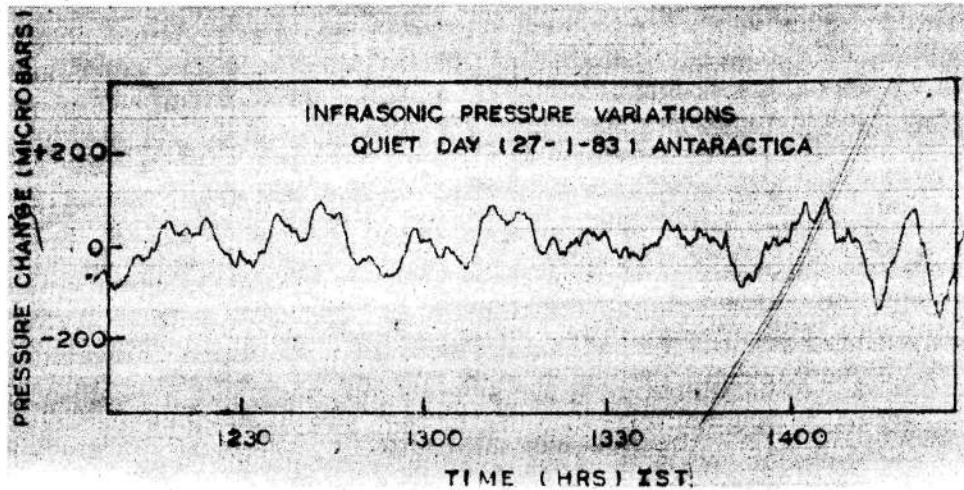


Fig. 1. Infrasonic pressure variation on a quiet day at Antarctica (27 Jan., 1983).

From our observations in India on Delhi and Hyderabad infrasonic pressure variations (Srivastava *et al.*, 1982), we had concluded that infrasonic waves generated in the Arctic and Antarctic auroral regions travel down to low latitudes under favourable conditions, with speeds of the order of 500 m/sec. We identified some sinusoidal patterns in the microbarograph, associated with the sub-storms. Microbarograph records at Antarctica were specially investigated for the magnetic bay effects. Whenever a sub-storm was recorded in the Hyderabad magnetograms, sinusoidal oscillations associated with sub-storms were noted in Antarctica microbarographs, with delay times in the range of 15–60 minutes. The geomagnetic latitude of our Antarctic station is 62°S, and it is well outside the auroral oval.

We had earlier (Srivastava *et al.*, 1982) interpreted (following Wilson, 1964) the sub-storm associated infrasonic waves as due to joule heating in supersonic auroral area during polar magnetic sub-storms. All the 6 sub-storm cases noted by us were in the same diurnal time zone, in the later afternoon and evening hours. As such the delay time could not be ascribed to diurnal location changes of the auroral oval (Akasofu, 1978). In the absence of information of exact location and size of the auroral oval during these events, the reason for the range in delay time observed, could not be inferred. One special case with the some unusual features is illustrated in Fig. 2 (a & b). An SSC was recorded at Hyderabad magnetogram at 1544 Hrs U.T. ($H = 49r$) on 9 January 1983. Sinusoidal oscillations, that we have associated with sub-storms, appeared in Antarctic microbarograph records 15 minutes later (Fig. 2a). It has been reported (Hirasawa, 1982) that an SSC observed in mid and low latitude can trigger sub-storm in auroral station. Fig. 2b shows some strong fluctuations in microbarograph records some 3 hrs prior to this event. This may or may not be related to the SSC event. Anyhow it is recorded here pending more such observations in future expeditions.

SOME SPECIAL EVENTS

The special nature and topography and location of Antarctica station and the fact that the daytime thermal disturbances observed in lower latitudes are absent, make the observation of special events and interesting features more convenient. We are reporting two specific cases where linkage of infrasonic pressure variation could exist with an earthquake and with some explosion or explosive movement.

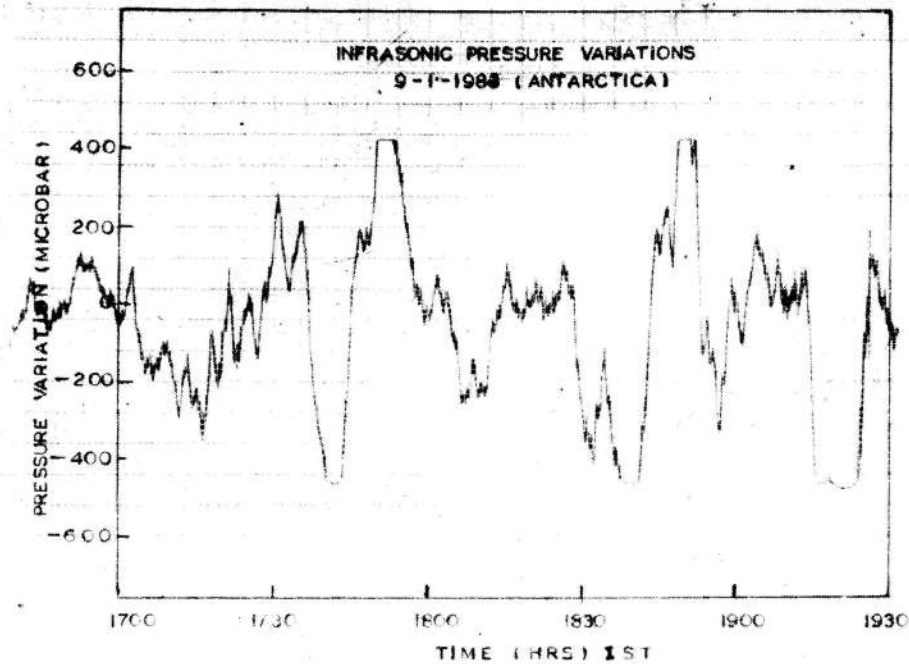


Fig. 2a. Sinusoidal oscillations in infrasonic pressure variation at Antarctica characteristic of a substorm (9 Jan. 1983).

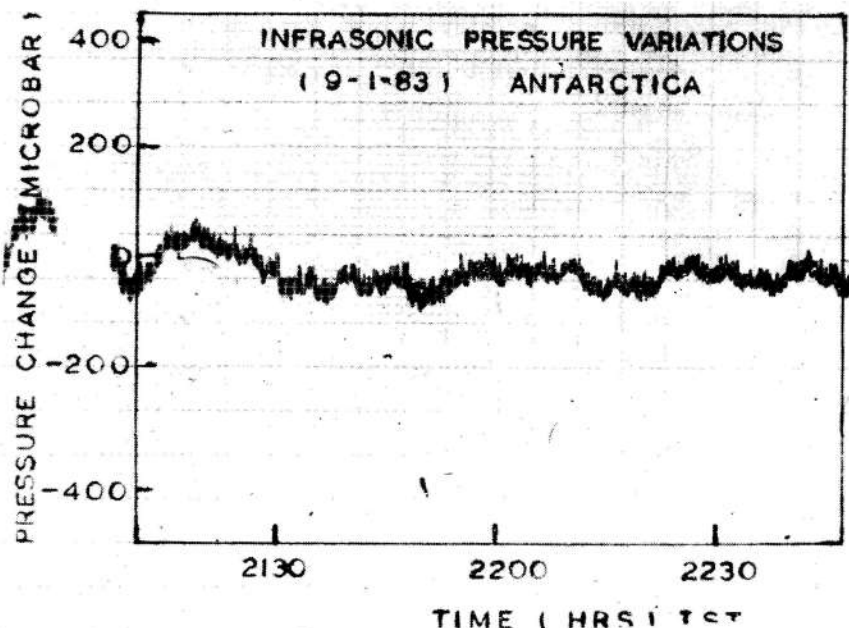


Fig. 2b. Strong fluctuations in infrasonic pressure variations prior to sub-storm (9 Jan. 1983).

(a) Earthquake effect

A major earthquake occurred at 2309 hrs. U.T. on 24 January, 1983 with the epicentre in the Andaman Sea (12.95°N , 93.64°E) at a depth of 79 km with a magnitude of 6.3 in Richter scale. This is shown as an effect in the magnetogram at Hyderabad.

Infrasonic Observations

Fig. 3 shows the microbarograph record at Hyderabad where a peak can be distinguished at 0640 hrs IST on 25 January, 1983, corresponding to a delay of about 2 hours which would be expected from a speed of 300 m/sec (expected acoustic speed in the troposphere).

The corresponding pattern (with a time delay of 4.5 hours) at Antarctica is also shown in the same figure. The wave form has apparently degenerated to a double pulse and the time delay for a distance (of the order of 7800 km) would correspond to a high speed of 500 m/sec. Perhaps, the propagation was through ocean waves. More observation of this phenomenon is needed to arrive at a definite conclusion.

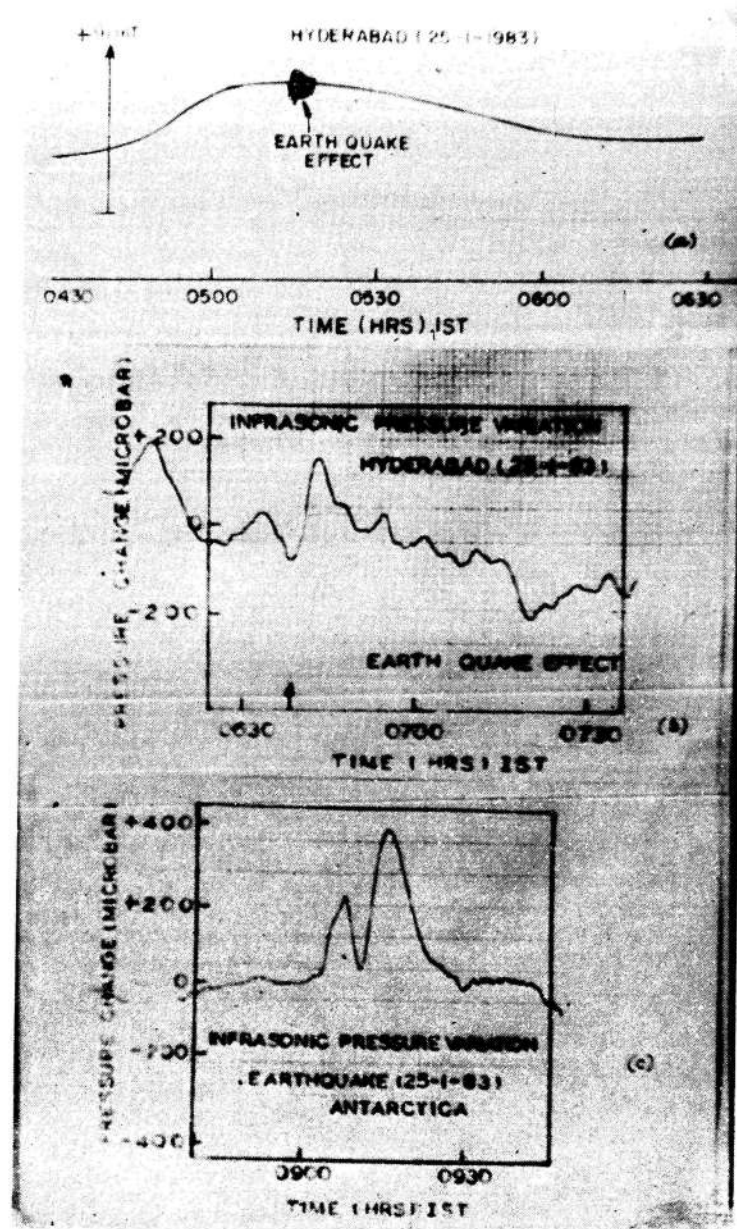


Fig 3a,b,c Earthquake effect recorded (a) on a Magnetogram at Hyderabad (25 Jan. 1983) and its effect (b) on infrasonic pressure variations at Hyderabad (25 Jan. 1983) and at (c) Antarctica (25 Jan. 1983).

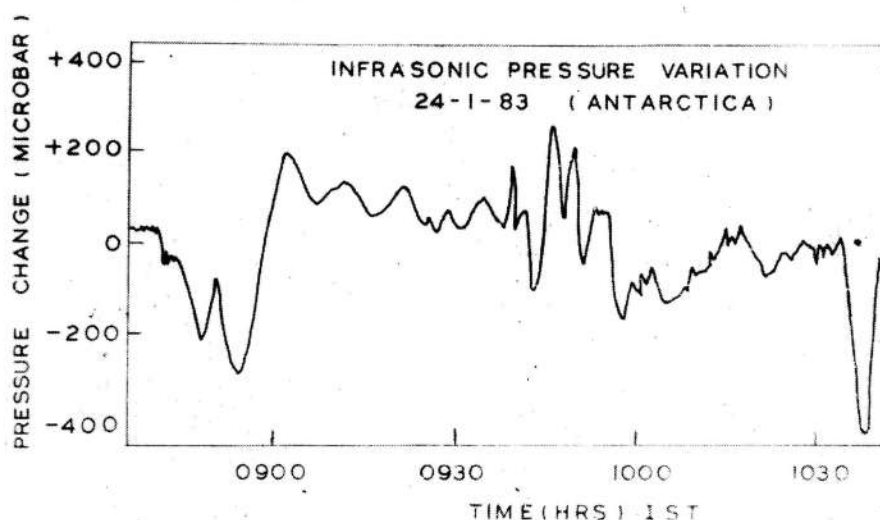


Fig. 4. A special pattern (Explosion ?) of infrasonic pressure variation at Antarctica (24 Jan. 1983).

(b) An explosion

Fig. 4 shows a special pattern observed in the Antarctica microbarograph records on the morning of 24 Jan., 1983. Two trains of waves of larger amplitudes are seen separated by a wave train of smaller amplitudes. The nature of the disturbance would indicate an explosive event, more likely on the seabed. In the absence of any specific information, it is not possible to speculate further. As the atmospheric conditions were calm and also similar to any other neighbouring day and no magnetic event was reported, atmospheric origin of this event is ruled out.

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REFERENCES

- Akasofu, S.I. (1978). *Upper Atmosphere Research in Antarctica*, edited by L.I. Lanzerotti and G.G. Park, American Geophysical Union, 157 pp.
- Campbell, W.H. (1967). In: *Physics of the Geomagnetic Phenomena*, edited by S. Matsushita, and H.W. Campbell, Academic Press, New York 860 pp.
- Hirasawa, T. (1982). *Proceedings of the IV Symposium on Co-ordinated Observations of the Ionosphere and the Magnetosphere in the Polar Regions*, edited by Takeshi Nagata, National Institute of Polar Research, Tokyo (Sept. 1982), 104 pp.
- Srivastava, B.J., R. Venkatachari and A.K. Saha (1982). *Geophys. J.R. Astr. Soc.*, 71, 241 pp.
- Venkatachari, R. and Bhartendu (1979). *Ind. J. Rad. Space Phys.*, 8, 273 pp.
- Wilson, C.R. (1969). *J. Geophys. Res.*, 74, 1812 pp.